Scientific Notations and Metric System

In science and engineering, we often work with numbers that are **extremely large** or **extremely small**.
For example:

- The **mass of a proton** (a tiny particle inside an atom) is:
  0.00000000000000000000000167 grams
- The number of **electrons passing in 1 ampere of current** every second is:
  6,250,000,000,000,000,000 electrons per second

These numbers are very hard to read, write, or calculate because of the many **zeros**. Imagine typing or writing that repeatedly, it's confusing and easy to make mistakes

To make this simpler, scientists developed a "shorthand" way of writing numbers called **scientific notation**.

Understanding Significant Digits (Significant Figures)

Before we dive into scientific notation, we need to understand **significant digits**, because they tell us **how accurate a measurement is**.

➤ What are Significant Digits?

Significant digits (or figures) are the **meaningful numbers** in a value that represent **actual measured or known quantities**. They show how precise the measurement or calculation is.

Example:

- If a car's weight is written as 3,000 pounds, we probably mean it's around that value, not exactly 3,000.
    - Only the "3" is significant.
    - The zeros are just placeholders (to show the scale of the number).
      1 significant figure
    - But if we write 3,005 pounds, the zeros are now meaningful because they show the number is measured more precisely.
      4 significant figures

In real-world measurements, we rarely know exact numbers. For example, when measuring voltage, resistance, or mass, our instruments have limits.
Significant digits tell us how confident we are about the value's accuracy.

What Is Scientific Notation?

Scientific notation is a compact way to express very large or very small numbers using **powers of ten**.
It keeps the **significant digits** in front and replaces all the **placeholder zeros** with a power of ten.

The General Form:

$N = a \times 10^n$

Where:

- **a** = a number between 1 and 10 (contains the significant digits)
- $\mathbf{10^n}$ = a power of ten that moves the decimal point

If **n** is **positive**, the number is **large** (move the decimal to the right).
If **n** is **negative**, the number is **small** (move the decimal to the left).

Expressing Very Large Numbers

Let's take the number of electrons flowing in one ampere of current:

6,250,000,000,000,000,000 electrons per second

Instead of counting zeros, we can write it as:

6.25×10^{18} electrons/second

- We moved the decimal point **18 places to the right**.
- The **significant digits** are **6.25**.
- The exponent **18** tells us how many times 10 multiplies itself.

So:

$10^{18} = 1{,}000{,}000{,}000{,}000{,}000{,}000$

This is much easier to read and calculate with!

Expressing Very Small Numbers

Now, let's look at the **mass of a proton**:

0.00000000000000000000000167 grams

To simplify:

1.67×10^{-24} grams

Here:

- We moved the decimal point **24 places to the right** to get 1.67.
- The exponent **of 24** shows the number is **very small**.
- The **negative sign** means move the decimal **to the left**.

Scientific notation makes even extremely small quantities readable and comparable.

Showing Accuracy Using Scientific Notation

Scientific notation doesn't just simplify numbers—it also helps show **how precise** a measurement is.

Let's revisit the **car weight example**:

Number (Normal)	Scientific Notation	Significant Figures	Meaning
3,000 lbs	3×10^3	1	Approximate (rounded)
3,005 lbs	3.005×10^3	4	Measured precisely
3,000 lbs (exact)	3.000×10^3	4	Exact to the nearest pound

Notice how **adding zeros after the decimal** in scientific notation shows precision — those zeros are **not placeholders**, but **significant** digits.

Working with Powers of Ten

Let's recall how powers of ten work:

Power	Meaning	Result
$\mathbf{10^0}$	$10^0 = 1$	1
$\mathbf{10^1}$	10×1	10
$\mathbf{10^2}$	10×10	100
$\mathbf{10^3}$	$10 \times 10 \times 10$	1,000
$\mathbf{10^{-1}}$	$1 \div 10$	0.1

10^{-2}	1 ÷ 100	0.01
10^{-3}	1 ÷ 1,000	0.001

Each step in **right** increases by ×10.
Each step in **left** divided by 10.

Understanding the Metric System

The **metric system** (also called the **SI system** is a **standardized system of measurement** used worldwide in science and engineering.

What makes it special is that it's based on powers of ten, the same idea behind scientific notation.

Instead of writing numbers with long strings of zeros, we can use prefixes to represent those powers of ten.
For example:

- 1,000 grams = **1 kilogram (1 kg)**
- 0.001 meter = **1 millimeter (1 mm)**

These prefixes make it much easier to handle very large or very small measurements.

Metric Prefixes and Powers of Ten

Each **prefix** corresponds to a **power of ten**, which means the **number of times ten is multiplied or divided**.

Below is a **number line** showing common prefixes used in science and engineering:

Prefix	Symbol	Power of Ten	Scientific Notation	Example
Giga	G	10^9	1,000,000,000	1 GW = 1,000,000,000 W
Mega	M	10^6	1,000,000	1 MHz = 1,000,000 Hz
Kilo	k	10^3	1,000	1 km = 1,000 m
(Base Unit)	—	10^0	1	1 m = 1 meter
Milli	m	10^{-3}	0.001	1 mm = 0.001 m
Micro	μ	10^{-6}	0.000001	1 μF = 0.000001 F
Nano	n	10^{-9}	0.000000001	1 ns = 0.000000001 s
Pico	p	10^{-12}	0.000000000001	1 pF = 0.000000000001 F

The Metric System as a Built-In Scientific Notation

The metric system is essentially scientific notation made simple.
Instead of writing "× 10^n" every time, you just use a prefix.

Example

6.25 × 10^3 → 6.25 kilograms (kg)
Both mean 6,250 grams.

So, prefixes are a **shortcut** for writing scientific notation in practical units.

Advantages of Scientific Notation

- Simplifies math and easier to multiply or divide large/small numbers
- Shows precision clearly
- Fewer zeros to write